

Eric Zhou

Software Engineer | Web & AI Applications

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EDUCATION

Queen's University

Bachelor of Computing (Honours)

– **GPA:** 4.1/4.3 (Dean's List)

Kingston, ON

Expected 2026

PROJECT EXPERIENCE

Semantic Code Search Engine | [GitHub](#) | *Python, FastAPI, Qdrant, Sentence Transformers*

Dec 2025

- Built AI code search using **Sentence Transformers** (384-dim embeddings) enabling natural language queries across **5 languages**.
- Implemented **hybrid search** combining Qdrant vector similarity with BM25 via Reciprocal Rank Fusion.
- Developed **Tree-sitter** AST parsing to extract functions/classes with signatures from Python, JS, TS, Go, Rust.

Travel Debate AI | [Live Demo](#) | [GitHub](#) | *Next.js 15, OpenAI, Redis*

Nov 2025

- Orchestrated **3 AI agents** with parallel execution (Promise.all), cutting response time by **50%** via **SSE streaming**.
- Architected modular backend with shared OpenAI/Redis clients and **React Error Boundaries** for graceful error recovery.
- Implemented **Redis** caching with connection pooling and **30-day TTL**, achieving **sub-50ms** retrieval for shareable links.

Repo Interview Prepper | [Live Demo](#) | [GitHub](#) | *React, Vite, Vitest, OpenAI API*

Nov 2025

- Implemented **real-time AI streaming** via SSE for mock interviews, reducing perceived latency by **70%**.
- Built **53 unit tests** with Vitest and React Testing Library; configured **GitHub Actions CI/CD** with **100% pass rate**.
- Architected codebase with **5 custom hooks**, serverless BFF pattern, and **rate limiting (20 req/min)** for API protection.

KingstonAccess – Accessible Parking Locator | [GitHub](#) | *React, TypeScript, Node.js*

Jan 2026

- Developed AI parking assistant using **LLM integration** for semantic NLP queries with **context-aware conversation memory**.
- Integrated **Google Maps Platform** APIs for accessible parking navigation; ranked **35+ lots** by **Haversine distance**.
- Built **WCAG 2.1 AA / AODA** interface with keyboard nav, ARIA, **bilingual (EN/FR)**, and **MFIPPA** privacy compliance.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Java, SQL, Bash

Frameworks: Next.js 15, React, Node.js, Express, FastAPI, Socket.io, Tailwind CSS

Databases & Tools: PostgreSQL, MongoDB, Qdrant, Redis, Docker, RabbitMQ, Git, Vitest

AI & ML: Sentence Transformers, Vector Embeddings, RAG Pipelines, LLM Integration, Prompt Engineering

WORK EXPERIENCE

Teaching Assistant - CISC 203 (Discrete Mathematics)

Sep 2025 – Present

Queen's University

Kingston, ON

- Mentored **200+ students** in Graph Theory and Combinatorics, translating abstract proofs into digestible logical steps.
- Analyzed recurring failure patterns in submissions to create “Common Errors” guide, reducing clarification inquiries by **15%**.
- Conducted root-cause analysis on student logic to identify edge cases in proofs, ensuring correctness across **50+ weekly assessments**.